

Quantum Chemistry of Atoms

"I always wanted to be somebody, but I should have been more specific"

1 INTRODUCTION

The classical theory is unable to explain the following experimental observations regarding small objects :

- (i) Discrete spectra emitted by excited atoms,
- (ii) Photoelectric effect,
- (iii) Variation of heat capacity of monoatomic solids with temperature, and
- (iv) Spectral distribution of energy in black-body radiation.

Max Planck was the first to suggest a new theory based on the quantum behaviour of energy. It was possible to account for the above experimental observations and hence a new theory was born, named, the *quantum theory*. In this chapter we will discuss quantum chemistry of atoms. The Schrödinger wave equation, wave function, particle in a box, hydrogen atom, Quantization of energy and angular momentum, Radial and angular parts of H-atom wave function/orbitals, and probability & charge distributions.

2 THE SCHRÖDINGER WAVE EQUATION

Consider an electron in an atom. Its total energy (E) is equal to the sum of kinetic energy and potential energy (V)

$$\begin{aligned} \Rightarrow E &= \frac{1}{2}mv^2 + V \\ E &= \frac{(mv)^2}{2m} + V = \frac{P^2}{2m} + V \\ &= \frac{P_x^2 + P_y^2 + P_z^2}{2m} + V \end{aligned} \quad \dots(1)$$

where $P = mv$ = momentum

This classical quantity E can be converted to the corresponding quantum mechanical operator for energy by the following replacements.

The operator corresponding to the energy E is called the *Hamiltonian operator* and is represented by the symbol H_{op}

(QC.1)

$$\begin{array}{l} \therefore \\ \text{Similarly,} \end{array} \left. \begin{array}{l} E \longrightarrow H_{op} \\ V \longrightarrow V \\ P_x \longrightarrow \frac{h}{2\pi i} \frac{\partial}{\partial x} \\ P_y \longrightarrow \frac{h}{2\pi i} \frac{\partial}{\partial y} \\ P_z \longrightarrow \frac{h}{2\pi i} \frac{\partial}{\partial z} \end{array} \right\} \dots(2)$$

From (1) and (2), we have

$$\begin{aligned} H_{op} &= \frac{1}{2m} \left\{ \left(\frac{h}{2\pi i} \frac{\partial}{\partial x} \right)^2 + \left(\frac{h}{2\pi i} \frac{\partial}{\partial y} \right)^2 + \left(\frac{h}{2\pi i} \frac{\partial}{\partial z} \right)^2 \right\} + V \\ &= \frac{-h^2}{8\pi^2 m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + V \\ \boxed{H_{op} = \frac{-h^2}{8\pi^2 m} \nabla^2 + V} \end{aligned} \dots(3)$$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is *Laplacian operator*.

For Hamiltonian operator, it is possible to write

$$\boxed{H_{op} \psi = E \psi} \dots(4)$$

This means when the H_{op} acts on the wave function ψ (the *eigen function*). It gives back the function (ψ) multiplied by a constant quantity E (the *eigen value* of the H_{op}).

From (3) and (4),

$$\left(\frac{-h^2}{8\pi^2 m} \nabla^2 + V \right) \psi = E \psi$$

or

$$\boxed{\nabla^2 \psi + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0} \dots(5)$$

The equation (5) is the required *Schrödinger equation*.

3 INTERPRETATION OF ψ and ψ^2

Schrödinger's equation is a second order differential equation and therefore has a large number of solutions. But only a few values of ψ have physical significance. These are those values of ψ which correspond to certain definite (or discrete) values of total energy called *eigen values*. The corresponding wave function is referred to as an *eigen function*.

The *wave function* (ψ) as such has no physical significance except that it represents the amplitude of the wave.

But ψ^2 is related to the probability of finding a particle with a definite value of energy within a certain domain in space.

THIS domain in space is between x and $x + dx$ for a one-dimensional problem ; an element of area for a two-dimensional problem ; and an element of volume for a three-dimensional problem.

Interpretation of ψ^2 by Schrödinger : In case of an electron, the value of ψ^2 at that any point around the nucleus gives a measure of the electronic charge density at that point.

Interpretation of ψ^2 by Max Born : According to this statistical interpretation of the electron wave function, the electron is still considered a particle and the value of ψ^2 at any point is the probability of finding the electron at that point at a given instant.

The probability amplitude implied that ψ must obey certain mathematical conditions.

The conditions which a wave function (ψ) must satisfy are : $\psi(x)$ must be single valued, finite and continuous for all physically possible values of x .

4 THE PARTICLE IN A BOX

The simplest application of the Schrödinger equation is to the problem of a particle moving in a box. For this, consider the motion of a particle of mass (m) along x -axis and confined between the walls of the container of length L (figure 1). Beyond L , the potential energy is infinitely high and there will be zero probability of finding the particle.

As $V(x) = \infty$ for $x < 0$ and $x > L$
 $\therefore \psi^2 = 0$ for $x < 0$ and $x > L$
 and thus $\psi = 0$ for $x \leq 0$ and $x \geq L$.

Its potential is zero between $x = 0$ and $x = L$ and rises abruptly to infinity as soon as it touches the walls.

But in the region $0 < x < L$, i.e., inside the one dimensional box, the particle is free and the potential energy V is taken as zero.

$$V(x) = 0 \text{ for } 0 \leq x \leq L$$

$\therefore$ The Schrödinger equation for the motion of the particle takes the form

$$\frac{d^2 \psi}{dx^2} + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0 \quad \dots(1)$$

$$\Rightarrow \frac{d^2 \psi}{dx^2} + \frac{8\pi^2 m}{h^2} E \psi = 0 \quad [\because V = 0] \dots(2)$$

Let $\frac{8\pi^2 m}{h^2} E = k^2$, so equation (2) becomes

$$\frac{d^2 \psi}{dx^2} + k^2 \psi = 0 \quad \dots(3)$$

Equation (3) is a differential equation whose general solution is

$$\psi = A \cos kx + B \sin kx \quad \dots(4)$$

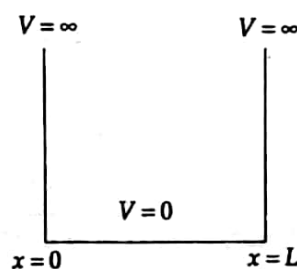


Fig. 1. A particle in a one dimensional region.

Proof. As $\psi = A \cos kx + B \sin kx$

$$\frac{d\psi}{dx} = A(-k) \sin kx + Bk \cos kx$$

$$\frac{d^2\psi}{dx^2} = A(-k^2) \cos kx + B(-k^2) \sin kx$$

$$= -k^2 (A \cos kx + B \sin kx)$$

$$\frac{d^2\psi}{dx^2} = -k^2 \psi \quad \text{or} \quad \frac{d^2\psi}{dx^2} + k^2 \psi = 0 \quad \text{Eqn. (3)}$$

As we have assumed that $\frac{8\pi^2 m}{h^2} E = k^2$

But $\hbar = \frac{h}{2\pi} \quad \therefore \quad \frac{h^2}{4\pi^2} = \hbar^2$

so $\frac{2mE}{\hbar^2} = k^2 \quad \Rightarrow \quad E = \frac{k^2 \hbar^2}{2m} \quad \dots(5)$

As $E = \text{kinetic energy} = \frac{1}{2} mv^2 = \frac{p^2}{2m} \quad \dots(6)$

From equations (5) and (6), we get

$$\frac{k^2 \hbar^2}{2m} = \frac{p^2}{2m}$$

$\Rightarrow \quad \boxed{P = \pm k \hbar} \quad \dots(7)$

Hence, the particle inside the box behaves as a free particle.

From equation (4),

$$\psi = A \cos kx + B \sin kx$$

At $x = 0$, equation (4) becomes

$$\psi = A \cos 0 + B \sin 0 = A$$

since at $x = 0$, $\psi = 0$ hence $\psi = A = 0 \quad \dots(8)$

Thus, equation (4) becomes

$$\psi = B \sin kx \quad \dots(9)$$

Imposing second boundary condition, $\psi \rightarrow 0$ for $x \rightarrow L$

$$\therefore \quad 0 = B \sin kL \quad \Rightarrow \quad \sin kL = 0$$

$$\Rightarrow \quad kL = n\pi \quad \Rightarrow \quad k = \frac{n\pi}{L} \quad \dots(10)$$

where n can have only integer values

i.e., $n = 1, 2, 3$

From equations (5) and (10), we get

$$E = \frac{k^2 \hbar^2}{2m} = \frac{(n\pi/L)^2 \hbar^2}{2m} = \frac{\left(\frac{n^2 \pi^2}{L^2} \frac{h^2}{4\pi^2} \right)}{2m}$$

$\Rightarrow \quad \boxed{E = \frac{n^2 h^2}{8mL^2} \quad n = 1, 2, 3, 4 \dots} \quad \dots(11)$

Since the constant n can have only integral values, it follows that the *energy* associated with the motion of a particle in a box is *quantized*.

In this case, n thus represents the *quantum number*. Because, at best $n = 1$ so lowest energy is $E_1 = \frac{h^2}{8mL^2}$, it is called as *zero point energy*. The fact that zero point energy (the lowest value of energy which the particle can possess) is not zero is consistent with the *uncertainty principle*. According to which, if the particle is confined to within a finite region, it ought to have some kinetic energy. Moreover, as the ψ is zero at the walls, but smooth, continuous and not zero everywhere, then it must be curved. This curvature in a wavefunction implies the possession of kinetic energy.

From equations (9) and (10), we get

$$\psi = B \sin \frac{n\pi}{L} x \quad \dots(12)$$

The above wavefunction contains the undetermined constant B . Its value can be found out by knowing that total probability of finding the particle from $x = 0$ to $x = L$ will be unity.

$$\text{Thus, } \int_0^L \psi \psi^* dx = 1 \quad \dots(13)$$

From equations (12) and (13), we get

$$\begin{aligned} & \int_0^L \left[B \sin \left(\frac{n\pi}{L} x \right) \right]^2 dx = 1 \\ \Rightarrow & B^2 \int_0^L \sin^2 \left(\frac{n\pi x}{L} \right) dx = 1 \\ \Rightarrow & \frac{B^2}{2} \int_0^L \left[1 - \cos \left(\frac{2n\pi x}{L} \right) \right] dx = 1 \\ \Rightarrow & \frac{B^2}{2} \left[(x)_0^L - \left\{ \frac{\sin \left(\frac{2n\pi x}{L} \right)}{\left(\frac{2n\pi}{L} \right)} \right\}_0^L \right] = 1 \\ \Rightarrow & \frac{B^2}{2} L = 1 \quad \Rightarrow \quad B = \sqrt{\frac{2}{L}} \quad \dots(14) \end{aligned}$$

From equations (7) and (10), we have

$$\begin{aligned} P &= \pm k \hbar = \pm \left(\frac{n\pi}{L} \right) \left(\frac{h}{2\pi} \right) \\ \Rightarrow & \boxed{P = \pm \frac{nh}{2L}} \quad \dots(15) \end{aligned}$$

From de-Broglie equation, $\lambda = \frac{h}{mv} = \frac{h}{p}$

$$\therefore \lambda = \frac{h}{nh/2L} = \frac{2L}{n}$$

$$\Rightarrow \boxed{L = \frac{n\lambda}{2}} \quad \dots(16)$$

Substituting the values of B from equation (14) in equation (9), we get

$$\boxed{\psi = \sqrt{\frac{2}{L}} \sin \frac{n\pi}{L} x} \quad \dots(17)$$

The energies, wave functions and probability densities of a particle in a box :

n	1	2	3	4	5
$E = \frac{n^2 h^2}{8mL^2}$	$\frac{h^2}{8mL^2}$	$4 \times \frac{h^2}{8mL^2}$	$9 \times \frac{h^2}{8mL^2}$	$16 \times \frac{h^2}{8mL^2}$	$25 \times \frac{h^2}{8mL^2}$

...(18)

Now, $\psi = \sqrt{\frac{2}{L}} \sin \frac{n\pi}{L} x$ [From equation (17)]

For $n=1$,

$$\boxed{\psi = \psi_1 = \sqrt{\frac{2}{L}} \sin \frac{\pi x}{L}}$$

Now at $x=0$,

$$\psi_1 = 0; \quad \psi_1^2 = 0$$

$$\text{at } x = \frac{L}{4},$$

$$\psi_1 = \frac{1}{\sqrt{L}}; \quad \psi_1^2 = \frac{1}{L}$$

$$\text{at } x = \frac{L}{2},$$

$$\psi_1 = \sqrt{\frac{2}{L}}; \quad \psi_1^2 = \frac{2}{L}$$

$$\text{at } x = \frac{3L}{4},$$

$$\psi_1 = \frac{1}{\sqrt{L}}; \quad \psi_1^2 = \frac{1}{L}$$

$$\text{at } x = L,$$

$$\psi_1 = 0; \quad \psi_1^2 = 0$$

...(19)

Similarly, for $n=2$,

$$\boxed{\psi = \psi_2 = \sqrt{\frac{2}{L}} \sin \frac{2\pi x}{L}}$$

$\therefore$ at $x=0$,

$$\psi_2 = 0; \quad \psi_2^2 = 0$$

$$\text{at } x = \frac{L}{4},$$

$$\psi_2 = \sqrt{\frac{2}{L}}; \quad \psi_2^2 = \frac{2}{L}$$

$$\text{at } x = \frac{L}{2},$$

$$\psi_2 = 0; \quad \psi_2^2 = 0$$

$$\text{at } x = \frac{3L}{4},$$

$$\psi_2 = -\sqrt{\frac{2}{L}}; \quad \psi_2^2 = \frac{2}{L}$$

$$\text{at } x = L,$$

$$\psi_2 = 0; \quad \psi_2^2 = 0$$

...(20)

etc.

With the help of equations (18), (19) and (20), we can draw the Fig. 2.

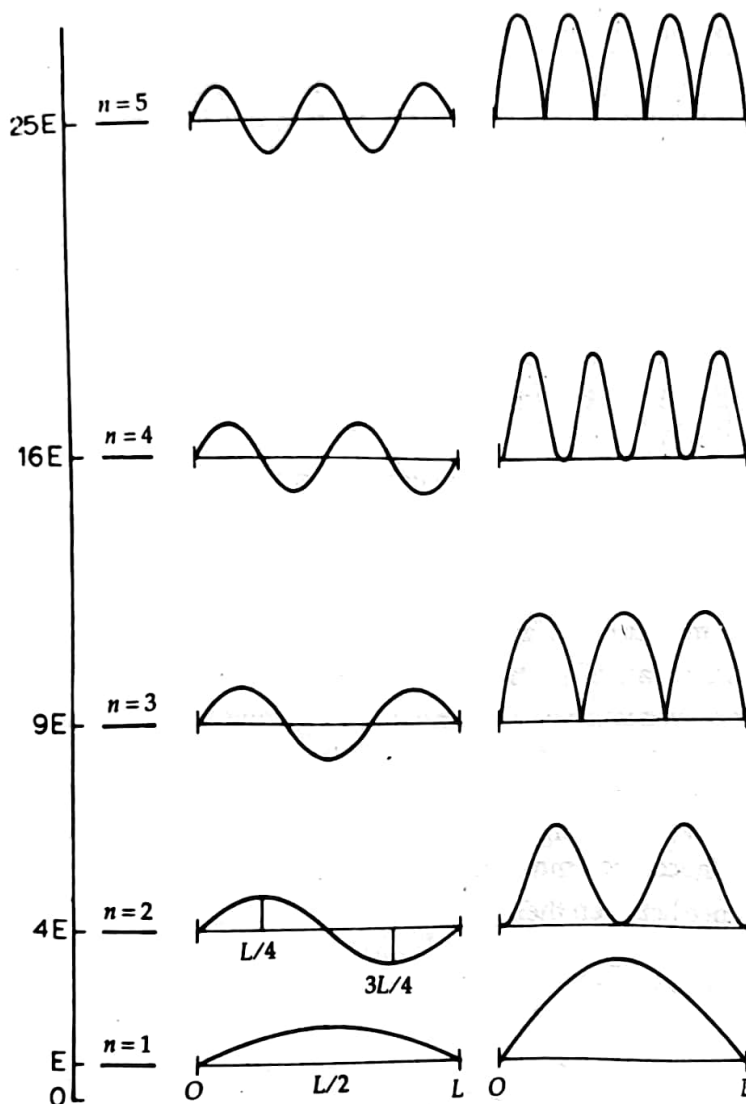


Fig. 2. The energies, wave functions and probability densities of a particle in a box.

The following points can be concluded :

(a) The wave functions besides being zero at the walls of the box also have zero values at various points within the box. The number of times, ψ becomes zero is equal to $(n - 1)$, where n is the quantum number. The point at which $\psi = 0$ is known as the *node of the wave function*.

(b) The number of nodes of the function and the energy associated with a wave function increases with n .

(c) The wave functions may be Symmetric or Antisymmetric w.r.t. the centre of the box.

Functions	Criteria	Examples of ψ from particle in a box
Symmetric or Even	$\psi(x) = \psi(-x)$	$\psi_1, \psi_3, \psi_5 \dots$
Antisymmetric or odd	$\psi(x) = -\psi(-x)$	$\psi_2, \psi_4, \psi_6 \dots$

(d) The probability density is not uniform at all positions. When n is small, probability density is more pronounced but when n is increased, the distribution becomes more and more uniform. For very high values of n , the distribution is completely uniform. Thus, classical behaviour (that on an average, the particle spends equal amounts of time at all points) emerges when n becomes very large. This fact is in agreement with the *correspondence principle*, which says that the predictions of quantum mechanics for a very large value of n are identical to those predicted by classical mechanics.

Calculation of ΔE between HOMO and LUMO of Polyene (An Application of particle in a Box) :

The π electron in polyene may be correlated with the system of particle in a one-dimensional box. For simplicity here we are assuming that polyenes are linear molecule although they are not. The length of the box may be taken as the end to end distance of the molecule which is further extended by a half C – C distance on either side of the molecule, i.e., another 154 pm distance is considered.

Thus knowing mass of electron (m_e) and length of 1 – D box (L), the electronic energy levels in polyenes can be determined using equation (11). The π -electrons of polyenes are distributed over these levels following the Pauli's exclusion and aufbau principles. The electron transition energy may be identified with the energy difference between the (n) {highest occupied molecular orbital (HOMO)} and the ($n + 1$) {Lowest Unoccupied molecular orbital (LUMO)}.

The difference between these two levels is

$$\begin{aligned}\Delta E = E_{n+1} - E_n &= \frac{(n+1)^2 h^2}{8m_e L^2} - \frac{n^2 h^2}{8m_e L^2} \\ &= \{(n+1)^2 - n^2\} \left(\frac{h^2}{8m_e L^2} \right) = (2n+1) \frac{h^2}{8m_e L^2} \\ \Rightarrow \boxed{\Delta E = (2n+1) \frac{h^2}{8m_e L^2}} &\quad \dots(21)\end{aligned}$$

Let us calculate ΔE in butadiene, $\text{CH}_2 = \overset{\text{H}}{\underset{|}{\text{C}}} - \overset{\text{H}}{\underset{|}{\text{C}}} = \text{CH}_2$

$\xleftarrow{\quad L \quad} \xrightarrow{\quad}$

$$\begin{aligned}L &= 1 \text{ (single bond length)} + 2 \text{ (double bond-length)} + 154 \text{ pm} \\ &= (154 \text{ pm}) + 2 (135 \text{ pm}) + 154 \text{ pm} = 578 \text{ pm} \\ &= 578 \times 10^{-12} \text{ m}\end{aligned}$$

$$n = \frac{\text{Number of electrons}}{2} = \frac{4}{2} = 2$$

$$\therefore \Delta E = (2n + 1) \frac{h^2}{8m_e L^2} \quad (\text{from equation 21})$$

$$= (2 \times 2 + 1) \frac{(6.626 \times 10^{-34} \text{ J s})^2}{8 (5.78 \times 10^{-10} \text{ m})^2 (9.11 \times 10^{-31} \text{ kg})} = 9.02 \times 10^{-19} \text{ J}$$

$$\Delta \tilde{E} = \frac{\Delta E}{hc} = \frac{9.02 \times 10^{-19} \text{ J}}{(6.626 \times 10^{-34} \text{ J s}) (3 \times 10^8 \text{ m s}^{-1})}$$

$$= 4.54 \times 10^6 \text{ m}^{-1} = 4.54 \times 10^4 \text{ cm}^{-1}$$

The corresponding experimental value is $4.61 \times 10^4 \text{ cm}^{-1}$.

From Fig. 2, in the $n = 2$ state, there is a finite (though varying) probability of finding the particle at all points on either side of the middle point of the box, but there is absolutely no chance of finding it in just the middle. Does it mean that the particle can move to and fro from one end to the other end without actually even touching the middle point? But precisely speaking, it is the conclusion based on quantum mechanical considerations. This can be explained by considering the particle to be a smeared out cloud of varying density with zero density at the middle of the box and maximum density on either side of it.

Let us now *verify the uncertainty principle for a particle in a 1 - D Box* :

$$P = \frac{nh}{2L}$$

$$\begin{aligned} \therefore \Delta P &= P_{n+1} - P_n \\ &= ((n+1) - n) \frac{h}{2L} = \frac{h}{2L} \end{aligned}$$

$$\text{If } L = \Delta x$$

$$\text{then } \Delta P \cdot \Delta x = \frac{h}{2}$$

$$\text{As } \frac{h}{2} \geq \frac{h}{4\pi}, \text{ so } \boxed{\Delta P \cdot \Delta x \geq \frac{h}{2\pi}}, \text{ thus uncertainty principle is verified.}$$

5 THE HYDROGEN ATOM

The hydrogen atom is simplest atom consisting of one electron and a proton of charge $+e$.

The time-independent form of Schrödinger equation is

$$\boxed{H_{op} \Psi_{\text{total}} = E_{\text{total}} \Psi_{\text{total}}} \quad \dots(1)$$

As the hydrogen atom has two particles *viz.*, proton and electron, thus the wave function (Ψ_{total}) depends on six co-ordinate variables, three each for proton in nucleus (x_n, y_n, z_n) and electron (x_e, y_e, z_e). The Hamiltonian operator consists of two terms, *viz.*, the kinetic energy and the potential energy terms. Moreover, the kinetic energy operator has two terms, one for the proton and the other for the electron.

Thus, we have

$$H_{op} = K.E_{op} + V_{op}$$

$$= \left(\frac{-h^2}{8\pi^2 m_e} \nabla_e^2 - \frac{h^2}{8\pi^2 m_n} \nabla_n^2 \right) - \frac{Ze'^2}{r} \quad \dots(2)$$

Here $Z = 1$ and in S.I. units $e' = \frac{e}{\sqrt{4\pi\epsilon_0}}$

The Schrödinger equation can then be written as

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{8\pi^2 m}{h^2} \left[E + \frac{e^2}{4\pi\epsilon_0 r} \right] \psi = 0 \quad \dots(3)$$

This equation can be solved by transforming the Cartesian coordinates to spherical coordinates r, θ, ϕ . These are defined in figure 3 below.

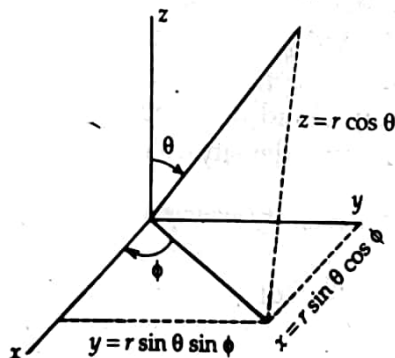


Fig. 3. Spherical polar coordinates.

The Schrödinger equation after transformation into spherical coordinates takes the form

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2} \sin^2 \theta \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2} \sin^2 \theta \frac{\partial^2 \psi}{\partial \phi^2} + \frac{8\pi^2 m}{h^2} \left(E + \frac{e^2}{r(4\pi\epsilon_0)} \right) \psi = 0 \quad \dots(4)$$

The wave equation (4) can also be written as a product of three functions the first of which depends only on r , the second only on θ , and the third only on ϕ .

$$\psi(R \Theta \Phi) = R(r) \Theta(\theta) \Phi(\phi) \quad \dots(5)$$

If we now substitute the value of ψ from equation (5) in equation (4), and solve it, three ordinary differential equations result. Each one of these differential equations on solving by imposing the boundary conditions lead to the emergence of three quantum numbers, which must be integers.

(a) *The principal quantum number (n)*, it arises from the freedom of the electron to vary its distance from the nucleus and have values 1, 2, 3 ...

(b) *The azimuthal quantum number or the angular momentum quantum number (l)*. It takes the integral values from 0 to $(n - 1)$.

The azimuthal quantum number describes the total angular momentum of the electron through the expression

$$L = \sqrt{l(l+1)} \frac{h}{2\pi}$$

It is customary to designate the values of l by letters which are derived from spectroscopic terms :

Value of l	0	1	2	3
Designation	s	p	d	f
Spectroscopic term	sharp	principal	diffuse	fundamental

(c) *The magnetic quantum number (m).* It takes the integral values from $-l$ to $+l$, including zero ; thus giving a total of $(2l+1)$ values.

The magnetic quantum number describes the z-component of the angular momentum of the electron through the equation

$$L_z = m \left(\frac{h}{2\pi} \right)$$

The eigen values for the energy of hydrogen or hydrogen-like atoms are given by

$$E_n = -\frac{2\pi^2 m e^4 z^2}{n^2 h^2 (4\pi \epsilon_0)^2}; \quad n = 1, 2, 3 \dots \quad \dots(6)$$

The E_n depends only on the value of n and not at all on l and m . Thus, all wave functions or orbitals with the same value of n but different values of l and m are degenerate, i.e., they have the same energy. For example, when $n = 2$,

$n = 2$	$l = 0$	$m = 0$
	$l = 1$	$\begin{cases} -1 \\ m = 0 \\ +1 \end{cases}$

Four-fold degenerate

The wave function ψ for electrons in atoms and molecules is called **atomic orbital**. The shape and orientation of an atomic orbital is mainly determined by l and m and is independent of n .

The 1s orbital

For the ground state, $n = 1$ and $l = 0, m = 0$. Such an orbital is called **1s orbital**. It depends only on the radial coordinate (r) and is independent of θ and ϕ . 1s orbital is spherically symmetrical in shape as shown in Fig. 4.

The normalized wave function for 1s orbital in the ground state is given below

$$\psi_{1s} = \frac{1}{\sqrt{\pi}} (Z/a_0)^{3/2} e^{-\sigma} \quad \dots(7)$$

where

$$a_0 = \frac{h^2 (4\pi\epsilon_0)}{4\pi^2 m e^2} \text{ and } \sigma = \frac{Z}{a_0} r$$

It is obvious from equation (7) that in 1s state the wave function and the probability distribution are independent of θ and ϕ .

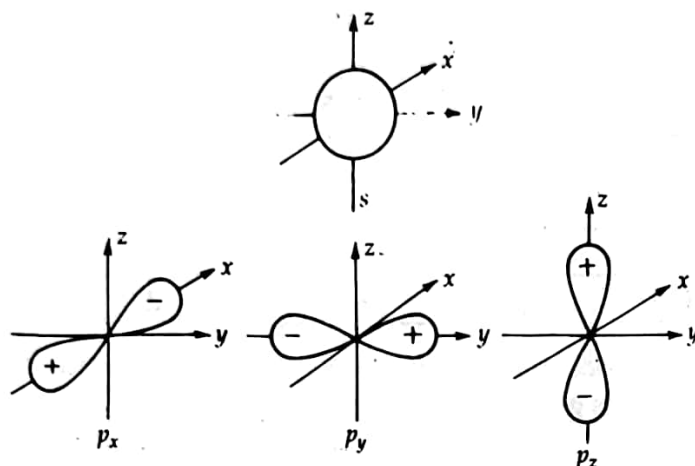


Fig. 4. Plots of the angular parts of the hydrogen atom wave functions for $l = 0$ (s) and $l = 1$ (p) orbitals.

The probability of finding the electron in space is given by

$$\Psi_{1s}^2 = \frac{1}{\pi} \left(\frac{z}{a_0} \right)^3 e^{-2\sigma} \quad \dots(8)$$

A plot of radial part $R(r)$ of hydrogen atom wave function for 1s and 2s orbitals are made as a function of r in Fig. 5(a).

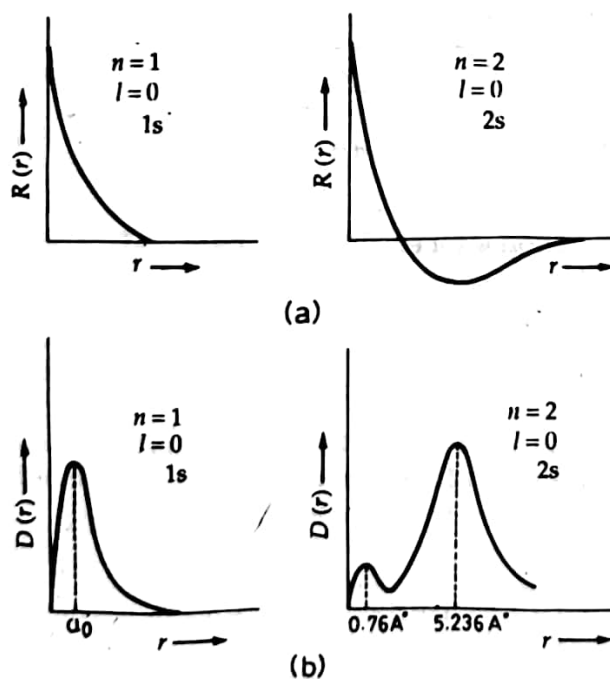


Fig. 5. (a) The variation of density, (b) probability density for finding the electron at a distance between r and $r + dr$.

The radial function always contains the factor $\exp\left(-\frac{Zr}{na_0}\right)$, where n is the principal quantum number. The radial wave function falls off more quickly with increasing r as the effective nuclear charge (Z) increases. The curve shows that $R(r)$ is maximum at $r = 0$, i.e., at the nucleus itself.

Radial distribution function $D(r)$

The probability of finding the electron in the region between two spheres of radii r and $r + dr$ is given by $D(r) = (4\pi r^2 dr) \psi^2$, where $4\pi r^2 dr$ is the volume of the spherical shell and ψ^2 is constant.

A plot of radial distribution function $D(r)$, versus r for 1s and 2s electrons are shown in Fig. 5(b). The curve shows a maximum at $r = a_0$, the radius of the first Bohr orbit.

It can be concluded that though the electron density is maximum at $r = 0$, the probability of finding the electron in a spherical shell near the nucleus is very small and decreases for both shorter and larger values than a_0 .

Difference between the Bohr atom and the quantum mechanical atom. In Bohr model of atom the electron always revolves in an orbit at a fixed distance from the nucleus whereas in the wave mechanical approach the electron is mostly located at this point, but there is a definite probability of finding it somewhere else also.

The 2s orbital

When $n = 2$ and $l = 0$, the orbital is the 2s orbital and it has the wave function

$$\psi_{2s} = \frac{1}{4\sqrt{2}\pi} \left(\frac{Z}{a_0}\right)^{3/2} (2 - \sigma) \exp\left(-\frac{\sigma}{2}\right)$$

A plot of the radial distribution function against r for 2s orbital gives two peaks as shown in Fig. 5(b). These peaks are main peaks and subsidiary peaks at a distance of $r = 5.2 a_0$ and $r = 0.8 a_0$ respectively. It means that 2s electron is most likely to occur at a distance of $5.2 a_0$ from the nucleus for most of the time. But there is a lesser probability of finding it at $r = 0.8 a_0$. It can be concluded that as the distance of the electron from the nucleus increases, more delocalization of the electron takes place and the electrostatic attraction between the electron and the nucleus decreases.

The 2p orbitals

When $n = 2$, l may be 1. In such a case we have p orbitals which have marked directional character because the wave function is dependent on θ and ϕ .

For states with $l = 1$ (p orbital); m can be $-1, 0, +1$. Different values of m correspond to orbitals of different orientations of the orbital angular momentum. The p -orbital with $m = 0$ has zero component of angular momentum about z -axis and is therefore called p_z orbital.

The normalized wave function for the $2 p_z$ orbital,

$$\psi_{2p_z} = \frac{1}{4\sqrt{2}\pi} \left(\frac{Z}{a_0}\right)^{3/2} \sigma^{-\sigma/2} \cos \theta$$

It has a region of greatest probability along the z -axis. It is apparent from the figure 4 that the probability of finding the electron in the xy plane is zero.

i.e., if $\theta = 90^\circ$, $\psi_{2p_z} = 0$.

The xy plane is thus nodal plane for $2p_z$ orbital. The other two orbitals viz., $2p_x$ and $2p_y$ have similar shapes but different orientations. As these orbitals are oriented along x and y axes respectively.

EXERCISES

1. What is ψ ? What informations are conveyed by ψ and ψ^2 ?
2. Write down the Schrödinger's wave equation. Explain the significance of the terms involved.
3. Consider a ball of mass 1 gm moving with a speed of 1 cm s^{-1} in a one dimensional box of edge length equal to 10 cm.
 - (i) Calculate its kinetic energy and the number n corresponding to this kinetic energy.
 - (ii) If the ball is to be promoted to the next higher quantum level, then how much of energy is required?
4. Prove that the energy associated with the motion of a particle in a one-dimension box is quantized.
5. Verify the Uncertainty principle for a particle in 1 - D box.
6. Write Schrödinger wave equation for hydrogen atom, explain the terms involved.
7. What is the difference between Bohr atom and the quantum mechanical atom?